Check for updates

*Journal of Biogeography*

WILEY

Journal of Biogeography

RESEARCH ARTICLE OPEN ACCESS

# From Rainforests to Open-Vegetation: The Biogeographic History of *Chamaecrista* (Leguminosae) in the Neotropics

Juliana Gastaldello Rando[1,2] | Eduardo Koerich Nery[3] | Anselmo Nogueira[3] | Matheus Martins Teixeira Cota[2] | Alexandre Zuntini[4] | Suzana Alcantara[5] | Gwilym Peter Lewis[4] | Domingos Cardoso[6] | Luciano Paganucci de Queiroz[2] | José Rubens Pirani[7]

[1]Programa de Pós Graduação Em Ciências Ambientais, Universidade Federal Do Oeste da Bahia, Barreiras, Bahia, Brazil | [2]Departamento de Ciências Biológicas, Herbário, Universidade Estadual de Feira de Santana, Feira de Santana, Bahia, Brazil | [3]Centro de Ciências Naturais e Humanas, Laboratório de Interação Planta-Animal, Universidade Federal Do ABC, São Bernardo do Campo, São Paulo, Brazil | [4]Accelerated Taxonomy Department, Royal Botanic Gardens, Kew, Richmond, UK | [5]Centro de Ciências Biológicas, Departamento de Botânica, Campus Trindade, Universidade Federal de Santa Catarina, Florianópolis, Santa Catarina, Brazil | [6]Diretoria de Pesquisas, Jardim Botânico Do Rio de Janeiro (JBRJ), Rio de Janeiro, Rio de Janeiro, Brazil | [7]Instituto de Biociências, Departamento de Botânica, Universidade de São Paulo, São Paulo, São Paulo, Brazil

**Correspondence:** Juliana Gastaldello Rando (ju.gastaldello.rando@gmail.com)

**Received:** 26 May 2025 | **Revised:** 6 November 2025 | **Accepted:** 15 December 2025

**Keywords:** caatinga | *Campos rupestres* | Cerrado | climatic shifts | open diagonal | savannas | seasonally dry forests

## ABSTRACT

**Aim:** We reconstructed the biogeographical history of a species-rich legume genus by testing alternative hypotheses on the origin and expansion of neotropical open-vegetation areas, with particular emphasis on the *campos rupestres*.

**Location:** Neotropics.

**Taxon:** *Chamaecrista* (Leguminosae), one of the most speciose genera of the neotropical region.

**Methods:** We reconstructed a fossil-calibrated phylogeny of *Chamaecrista* species and built a comprehensive database of species occurrences. We fitted and compared nine biogeographic models that differed in area availability and dispersal rates across time frames. These time frames were defined based on major climatic processes that have influenced neotropical open-vegetation areas.

**Results and Main Conclusions:** Under the best-fitting biogeographical model, *Chamaecrista* originated in the Amazon during the Mid Eocene. A single ancestral *Chamaecrista* lineage colonised the *campos rupestres* before the Late Oligocene–Mid Miocene, when climatic shifts presumably expanded lowland open-vegetation areas in the Neotropics. Most diversification of *Chamaecrista* species occurred in the *campos rupestres* from the Late Miocene to the Quaternary. Multiple ancestral *Chamaecrista* lineages dispersed from the *campos rupestres* to the Cerrado from the Mid Miocene onwards, while few lineages dispersed into the Caatinga during the same period. We propose that the *campos rupestres* might have functioned as a montane reservoir of plant diversity, harbouring lineages that later dispersed into adjacent lowland areas.

## 1 | Introduction

The Neotropics harbours about 110.000 species of seed plants (Antonelli and Sanmartín 2011), many endemic and highly threatened (Myers et al. 2000; Bachman et al. 2024). This outstanding plant species diversity has been the subject of multiple studies investigating its tempo of evolution and historical correlates (Hughes et al. 2013). These studies have highlighted major geological and climatic processes shaping neotropical plant diversity (Hughes et al. 2013; Antonelli 2022), such as the Andean mountain uplift (Pérez-Escobar et al. 2022), the Miocene climatic shift (Steinthorsdottir et al. 2021), and the

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Journal of Biogeography* published by John Wiley & Sons Ltd.

Panama Isthmus formation (Leigh et al. 2013). These historical processes have been largely supported by biogeographic studies on plant lineages centered in the Amazon, Andes, and Atlantic Forest (Antonelli et al. 2009; Fine et al. 2014; Bünger et al. 2016). In contrast, few biogeographic analyses have addressed these historical processes in open-vegetation areas (Simon et al. 2009; Simon and Pennington 2012; Barros et al. 2020; Vasconcelos et al. 2020; Fernandes et al. 2022). Such an investigation bias precludes a more comprehensive understanding of neotropical plant diversity and biogeography (Hortal et al. 2015), also enforcing current conservation gaps for open-vegetation areas (Fonseca and Venticinque 2018).

The *campos rupestres* are mosaics of montane open grassy vegetation and rock outcrops occurring over the Eastern Brazilian highlands (Schaefer et al. 2016), whose formation can date back to the Paleoproterozoic (2.50–1.75 Ga) (Alkmim 1998; Pedreira and de Waele 2008). The *campos rupestres* are primarily found 900 m above sea level (m a.s.l.), forming enclaves within the Atlantic Forest, Cerrado, and Caatinga biogeographic dominion (Alves et al. 2014). Throughout their occurrences, the *campos rupestres* are under seasonally dry climates and have nutrient-impoverished soils, with some areas also subject to natural fires (Silveira et al. 2016). Under these abiotic conditions, the *campos rupestres* hold approximately 5.500 species of vascular plants (Flora e Funga do Brasil 2025), with great species richness of Asteraceae, Cyperaceae, Fabaceae, Poaceae, and Velloziaceae (Conceição et al. 2016). The *campos rupestres* have widely been treated as part of the Cerrado dominion, given their ecological and floristic similarities (Neves et al. 2018). However, a recent regionalization proposal has recognised the *campos rupestres* as a distinct biogeographic unit based on their high levels of plant endemism (Colli-Silva et al. 2019). Hence, the *campos rupestres* gained renewed attention as a conservation priority, but their biogeographic history has long been a matter of debate.

Neotropical open-vegetation areas presumably originated with climatic shifts from the Late Paleogene onwards (Jaramillo 2023). Firstly, temperatures slowly increased in the Southern hemisphere during the Late Oligocene, but rapidly dropped in the Oligocene-Miocene transition (O'Brien et al. 2020). Secondly, drier climates became more pronounced in the Southern hemisphere from the Late Miocene (Strömberg 2011; Steinthorsdottir et al. 2021). These climatic processes presumably promoted the expansion of open-vegetation areas over rainforests, forming a contiguous region from northeastern Brazil to northern Argentina, i.e., the open diagonal (Guillory et al. 2024). The open diagonal currently includes three biogeographic dominions: the Caatinga seasonally dry tropical forests and woodlands, the fire-prone tropical savannas of the Cerrado, and the Chaco subtropical dry woodlands (Werneck 2011). These open-vegetation dominions differ with respect to geology, soils, and disturbance regimes (Pennington et al. 2000; Werneck 2011), but all of them presumably accumulated their plant species diversity over the climatic shifts between the Late Oligocene and Mid Miocene (Jaramillo 2023). However, the *campos rupestres* might have accumulated their plant richness by distinct processes. Phylogenetic studies suggest that several plant lineages endemic to the *campos rupestres* originated relatively recently (~5 Mya) (Vasconcelos et al. 2020). These recent plant radiations would have derived from a pronounced in situ speciation during the glacial–interglacial cycles in the Plio-Pleistocene (Vasconcelos et al. 2020). Alternatively, ecological studies suggest that plant lineages endemic to the *campos rupestres* originated from edaphic specialisation and long-term persistence. The edaphic specialisation may have been promoted by the edaphic heterogeneity in the *campos rupestres*, while long-term persistence might have resulted from a greater climatic stability in these mountain regions (Silveira et al. 2016). Therefore, the *campos rupestres* might have accumulated their endemic flora by biogeographic processes distinct from those of other neotropical open-vegetation areas.

*Chamaecrista* (L.) Moench (subfamily Caesalpinioideae) is the second most species-rich legume genus in neotropical open vegetations, surpassed only by *Mimosa* L. (Dutra et al. 2009; Rando et al. 2020). The genus comprises 368 species worldwide, approximately 180 occurring exclusively in the Cerrado, including the *campos rupestres* (Rando et al. 2020, 2024). *Chamaecrista* species are woody plants with varying life forms (trees, shrubs, or subshrubs), occupying rainforest and open-vegetation areas (Rando et al. 2024). *Chamaecrista* species are primarily buzz-pollinated by large bees, such as bumblebees and carpenter bees (e.g., Gottsberger and Silberbauer-Gottsberger 1988; Barbosa et al. 2023). These pollinators can facilitate long-distance pollen transfer, as observed in *C. fasciculata* (Fenster 1991). In contrast, most *Chamaecrista* species possess elastically dehiscent pods (Rando et al. 2024) that disperse seeds over relatively short distances—typically limited to only two or three meters, based on observations in *C. fasciculata* (Lee 1984; Fenster 1991). Consequently, this restricted seed dispersal likely limits the geographic expansion of *Chamaecrista* populations (Fenster et al. 2003). Moreover, all analysed *Chamaecrista* species so far can form mutualistic symbiosis with nitrogen-fixing bacteria (Casaes et al. 2024), possibly under a strict dependence of these bacteria (e.g., Souza et al. 2024, 2025). Also, many *Chamaecrista* species have extra-floral nectaries (Alencar et al. 2013; Rando et al. 2024), which allow ant-mediated defence against herbivores (e.g., Rutter and Rausher 2004; Chinarelli et al. 2022). Throughout their geographic distribution, *Chamaecrista* species have evolved distinct adaptive responses to environmental conditions in neotropical open-vegetation areas, such as the developed root system (Prochazka et al. 2024). Therefore, *Chamaecrista* represents an excellent model for investigating the biogeographical dynamics between the Cerrado and the *campos rupestres*.

Our study investigates the biogeographical history of the genus *Chamaecrista* to explore how ancient geological and climatic processes have shaped its species diversity across neotropical open-vegetation areas, particularly the Cerrado and the *campos rupestres*. We reconstructed the most comprehensively sampled phylogenetic tree of *Chamaecrista* and its closest relatives and defined their biogeographical units based on a taxonomically curated dataset of species occurrence. Finally, we fitted and compared distinct biogeographical models to infer when *Chamaecrista* species have colonised open vegetations in the Neotropics, evaluating alternative hypotheses for the colonisation of the *campos rupestres*.



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

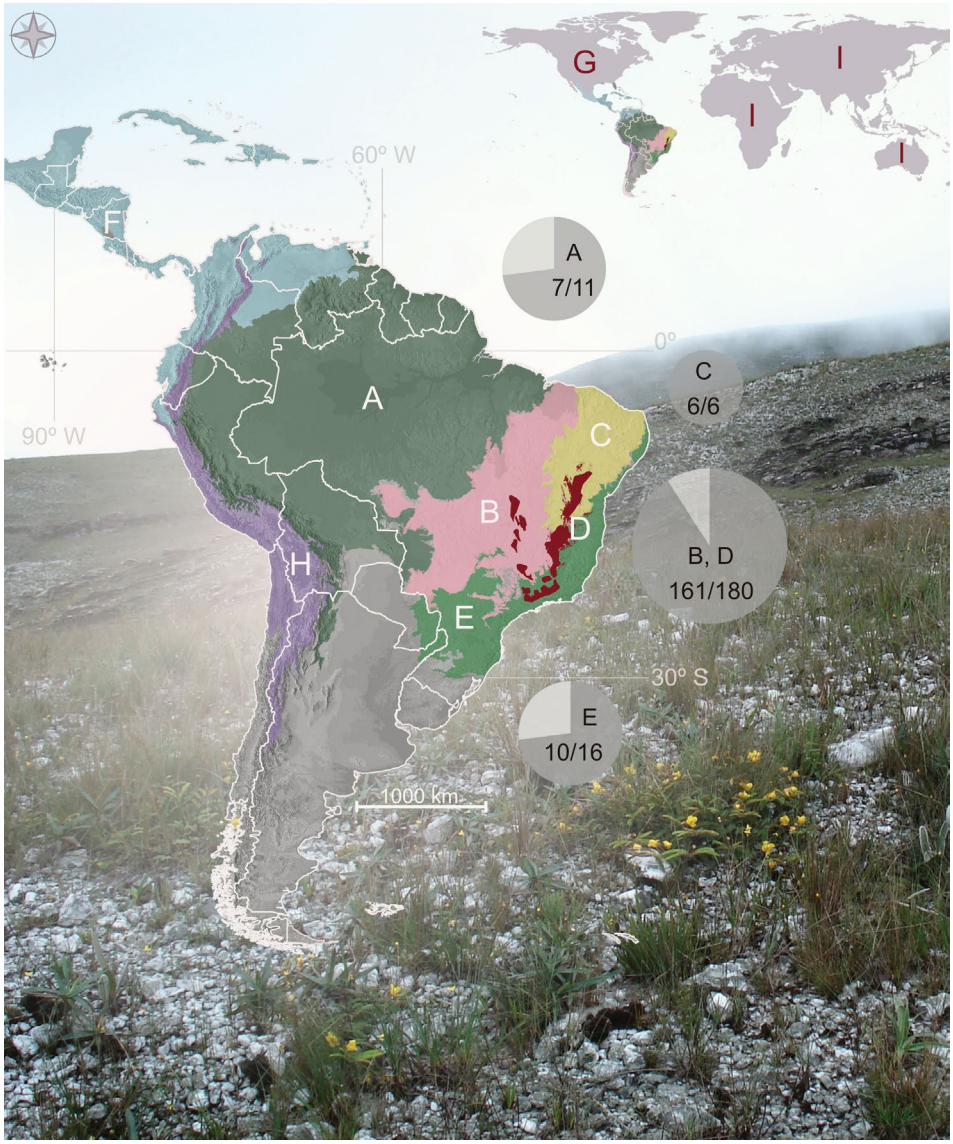



**FIGURE 1** | The nine biogeographic areas considered in our study (see text for a detailed explanation). The biogeographic areas in the Neotropics are depicted in the larger map, while areas outside the Neotropics are depicted in the smaller map, at upper right corner. The pie charts display the sampled/total number of *Chamaecrista* species in each biogeographic area, demonstrating our extensive sampling effort over open-vegetation areas. These numbers do not include species that are widely distributed. Each biogeographic area is labelled by letter: (A) Amazon; (B) Cerrado; (C) Caatinga; (D) *campos rupestres*; (E) Atlantic Forest; (F) Central America; (G) North America; (H) Andes; (I) Extra-American. The background photo shows the narrowly endemic, yellow-flowered *Chamaecrista olesiphylla* (Vogel) H.S.Irwin & Barneby thriving in a typical *campos rupestres* setting at Serra Talhada mountain in Congonhas do Norte, Minas Gerais state. Photo: Juliana Gastaldello Rando.

## 2 | Material and Methods

### 2.1 | Taxon Sampling

For the phylogenetic analysis, we sampled molecular data from *Chamaecrista* species and closely related lineages, following the current classification (Rando et al. 2024). *Chamaecrista* is a pantropical genus comprising 368 species, of which 295 are restricted to the Neotropics (Rando et al. 2024). Our molecular dataset included 231 *Chamaecrista* species, representing 63% of the genus and 73% of the species occurring in the Neotropics, and 184 endemic species to Brazil (Figure 1). We also included 50 species from other genera belonging to the tribe Cassieae, proportionally representing each genus based on its species diversity: 37 out of 287 species of *Senna*, 7 out of 39 species of *Cassia*, all three species of *Vouacapoua*, 1 out of 3 species of *Recordoxylon*, and the single species of the monotypic genera *Batesia* and *Melanoxylon*.

For each species, we gathered Sanger-sequence data from five loci: the nuclear ITS and ETS, and the plastidial *trnD-T*, *trnL-F*, and *trnK-matK*. We retrieved 910 sequences from GenBank, preferably selecting from accessions generated by experts in Leguminosae, most of which were from our own previously published phylogenetic studies (Conceição et al. 2009; Rando et al. 2016; LPWG 2017; Souza et al. 2021; Cota et al. 2025). The compiled number of sequences for species, as well as the complete list of vouchers with associated GenBank accession numbers, is presented in the Data S1 and Table S1a,b. We also generated 47 new sequences for all loci, following the same



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

DNA extraction, PCR amplification, and sequencing protocols from other phylogenetic studies on *Chamaecrista* (Conceição et al. 2009; Rando et al. 2016; Queiroz and Snak 2020). The detailed laboratory procedures are described in the Data S1.

## 2.2 | Phylogenetic Reconstruction

Sequences from the same locus were aligned using the G-INS-i algorithm in MAFFT (Katoh and Standley 2013), and alignments were manually edited to remove poorly aligned regions, resulting in a final dataset of the combined five nuclear and plastid loci comprising 5250 bp. Substitution patterns and interlocus congruence were assessed using the R package *phangorn* (Schliep et al. 2017). We compared 31 substitution models for each locus by using the Akaike Information Criterion (AIC; Akaike 1974) (Data S1 and Table S1a). We assessed congruence by generating 100 maximum likelihood (ML) bootstrap trees for each locus and comparing their topology. We compared the topology of ML bootstrap trees by a principal coordinate analysis (PCoA) on their Robinson and Foulds' distance (Robinson and Foulds 1981). We found that the *matK* and ITS loci were considerably incongruent with the others (Data S1 and Figure S1a).

To minimise the impact of gene incongruence, we applied concatenation to infer a minimal congruent topology for the main phylogenetic inference. Concatenation amplifies the common phylogenetic signal across genes (McVay and Carstens 2013), averaging down the noise signal from the incongruent genes (Gadagkar et al. 2005). We concatenated our five alignments and assessed their joint substitution model as previously described. We ran a maximum-likelihood (ML) phylogenetic analysis over the concatenated alignment by utilising the IQ-Tree algorithm (Nguyen et al. 2015), assuming the most dissimilar sequences as the outgroup (*Batesia floribunda* + *Vouacapoua* spp.). We measured tree node support by 1.000 bootstrap iterations. We retrieved the ML phylogenetic tree and collapsed all tree nodes with less than 90% bootstrap support. The remaining high-support nodes formed the minimal congruent topology for the time-calibrated phylogenetic inference (Data S2).

A time-calibrated phylogenetic tree was reconstructed with BEAST 2.5 (Bouckaert et al. 2019). We set separate substitution models for each locus. We assumed a Random Local Clock (RLC) to accommodate different shifts of substitution rates further from calibration nodes (Crisp et al. 2014), and a Birth-Death model for tree topology with two calibration nodes: (1) the most recent common ancestor (MRCA) of the Cassieae tribe (Normal distribution, mean: 53 Mya, sd: 1.567 mya) based on previous estimates (Bruneau et al. 2008); and (2) the MRCA of *Senna* (Log Normal distribution, mean: 45 Mya, sd: 0.005) based on the fossil record (Herendeen et al. 1992). Moreover, we constrained phylogenetic reconstruction based on the minimal congruent topology previously described. We implemented adaptive MCMC chains to perform a faster search in the phylogenetic tree space (Müller and Bouckaert 2020). We ran four MCMC chains with $3 \times 10^7$ iterations, sampling every $3 \times 10^4$ iterations. We verified convergence of all chains in Tracer 1.7 (Rambaut et al. 2018), observing that every parameter achieved an effective sample size (ESS) > 200. After discarding the first 20% of each run as burn-in, we summarised parameter estimates on the Maximum Clade Credibility (MCC) phylogenetic tree. The MCC phylogenetic tree was similar to gene trees from congruent loci, thus with little influence from gene incongruence (Data S1).

## 2.3 | Species Distribution and Biogeographic Regionalization

To assign the distribution of all neotropical *Chamaecrista* species to the biogeographic areas, we extracted species occurrences from the Global Biodiversity Information Facility (GBIF) using the R package *rgbif* (Chamberlain et al. 2025) (GBIF DOI in Data S3). We gathered occurrences for 324 *Chamaecrista* species, comprehensively representing the genus diversity. Importantly, most GBIF occurrence records in our dataset derive from *Chamaecrista* specimens that the lead author has meticulously curated and taxonomically revised over the past 15 years, ensuring a uniformly vetted and reliable identification framework. We filtered out occurrences without coordinates, those with erroneous placement, or deriving from historical collections (here considered before 1950) by utilising the R package *CoordinateCleaner* (Zizka et al. 2019). After filtering, we checked for occurrences outside the known geographic distribution of each *Chamaecrista* species. When feasible, we verified the taxonomic identification of these outlier occurrences via digital herbaria, correcting those with erroneous identification.

After checking the data, we adopted a network-based approach to delimit biogeographic areas (Vilhena and Antonelli 2015). This method applies network theory to simplify and highlight the underlying patterns of species distributions, making the bioregionalization less subjective than relying solely on expert opinion (Edler et al. 2016). This method was implemented using the Infomap Bioregions interactive web application, which processes species distribution data to delineate bioregion maps and identify taxon-specific biogeographic areas (Edler et al. 2016), following the parameters detailed in Data S3. Subsequently, we analysed the areas generated by Infomap and identified equivalent, previously classified regions in order to name the areas obtained. We named our biogeographic areas after the Brazilian biomes defined by IBGE (IBGE 2025) and the biogeographic regionalization proposed by Morrone et al. (2022). As a result, we recognised nine taxon-specific biogeographic areas: Amazon (A), Cerrado (B), Caatinga (C), *campos rupestres* (D), Atlantic Forest (E), Central America (F), North America (G), Andes (H), and extra-America (I) (Figure 1). We considered the Andes as a biogeographic area because some *Senna* species occur in this dominion, despite its negligible diversity of *Chamaecrista* species. Finally, we assigned *Chamaecrista* and allied species to these biogeographic areas based on their recorded occurrences (Data S4), which we cross-validated against their known geographic ranges using published distribution data, herbarium records, and our own field-collected observations.

## 2.4 | Biogeographic Analyses

To investigate the biogeographic history of *Chamaecrista*, we utilised three algorithms implemented in the 'BioGeoBEARS' package (Matzke 2013). All algorithms represent biogeographic evolution as a function of dispersal (d), local extinction (e), and



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

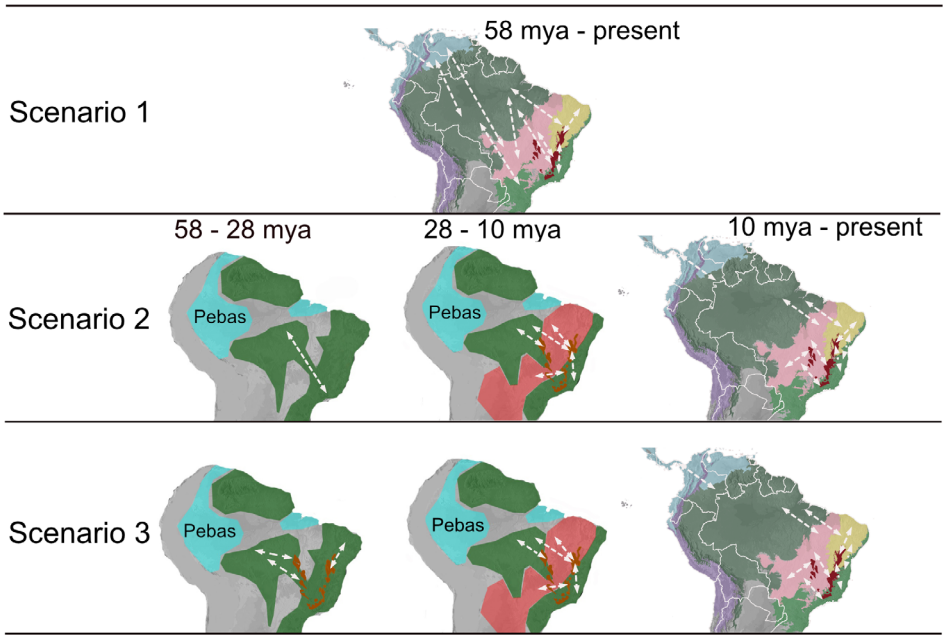



**FIGURE 2** | The three biogeographic scenarios considered in our analyses: Scenario 1, the *campos rupestres* and other areas are equally available throughout time, dispersal is unconstrained among areas. Scenario 2, the *campos rupestres* and other open-vegetation areas become available from the Late Oligocene—Mid Miocene (28–10 Mya) onwards, dispersal is constrained among non-neighbour areas. Scenario 3, the *campos rupestres* is available before the Late Oligocene—Mid Miocene (28–10 Mya), preceding lowland open-vegetation areas, and dispersal is constrained among non-neighbour areas. The Late Oligocene—Mid Miocene is considered a period for the contraction of rainforest and the initial formation of open-vegetation areas. Throughout, colours indicate presumed vegetation types in past time frames: Rainforests, deepgreen; lowland open-vegetation, lightred; *campos rupestres*, darkred; lake systems, cyan, transition areas, grey. Illustrated maps based on Antonelli et al. (2009) and additional information from Gottsberger and Silberbauer-Gottsberger (2006).

narrow sympatric speciation (y), but each algorithm considers a different set of additional processes. The Dispersal-Extinction-Cladogenesis (DEC) algorithm additionally considers narrow vicariance (v) and asymmetrical sympatric speciation (s). The Dispersal-Vicariance (DIVALIKE) algorithm additionally considers narrow and widespread vicariances (v). The Bayesian Biogeographic Analysis (BAYAREALIKE) algorithm additionally considers widespread sympatric speciation (y). Throughout, we did not consider founder events (j) that can model rapid changes of distribution without intermediate states (Matzke 2014), since *Chamaecrista* species have a short-distance dispersal, likely favouring gradual over rapid geographic expansion. All algorithms considered the nine biogeographic areas previously delimited for *Chamaecrista* species, but they allowed ancestral species to occupy up to eight areas simultaneously, as observed in extant species. This restriction was also needed to alleviate running time for ancestral area reconstruction.

We considered three biogeographic scenarios that varied with respect to area availability and dispersal rates across time frames (Figure 2). Because the Late Oligocene and Early and Mid Miocene have been regarded as the onset of open-vegetation areas in the Neotropics (Jaramillo 2023), we considered three time frames for our biogeographic models: (1) Eocene to Early Oligocene (58–28 Mya); (2) from Late Oligocene to Mid Miocene (28–10 Mya); and (3) from Late Miocene to Quaternary (10 Mya—present). Under the first scenario, areas were equally accessible and dispersal was unconstrained across time frames. Under the second scenario, the open-vegetation areas (B, C and D) became available only during the second time frame, and dispersal rates were reduced (0.5) among non-neighbour areas across all time frames. Under the third scenario, the *campos rupestres* (D) became available during the first time frame, preceding the other open-vegetation areas (B and C) during the second time frame, and dispersal rates were reduced (0.5) among non-neighbour areas across all time frames. For the second and third scenarios, Central America (F) became available during the third time frame, when the Panama Isthmus would have formed and facilitated plant dispersal (Leigh et al. 2013). Moreover, we reduced dispersal rates (0.1) to the extra-American area (I) because trans-oceanic dispersal processes are presumably more rare than continental ones (Data S5).

We fitted all three algorithms, DEC, DIVALIKE and BAYAREALIKE, considering all three biogeographic scenarios, resulting in nine alternative models. Beyond the biogeographic history of neotropical *Chamaecrista*, each model represented a hypothesis on how the *campos rupestres* had accumulated its outstanding plant endemism. Models considering the first biogeographic scenario represent a hypothesis where the *campos rupestres* would have gradually accumulated plant lineages, without an effect from the climatic shifts of the Late Oligocene and Mid Miocene. Models considering the second biogeographic scenario represent a recent colonisation hypothesis where the *campos rupestres* would have accumulated plant lineages over the climatic shifts of the Late Oligocene and Mid Miocene, together with other open-vegetation areas in the Neotropics. Models considering the third biogeographic scenario represent an ancient colonisation hypothesis where the *campos rupestres* would have gradually accumulated plant lineages before the



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

climatic shifts of the Late Oligocene and Mid Miocene, preceding lowland open-vegetation areas in the Neotropics. We compared the nine alternative models by the Akaike Information Criterion (AIC) (Akaike 1974), choosing the lowest score as the best-fitting model. Additionally, we utilised the Akaike weight (AICw) to assess the conditional probabilities of the nine models (Burnham and Anderson 2002).

## 3 | Results

### 3.1 | Phylogenetic Relationships and Divergence Times

Our phylogenetic analysis strongly supported (posterior probability, pp. = 1.0) the monophyly of *Chamaecrista*, with the clade *Cassia* + *Senna* as its sister group (Figure 3). *Chamaecrista* diverged from its sister group about 54.2 mya (95% High Probability Density, HPD = 57.4–51.3 mya), later diversifying about 41.2 mya (95% HPD = 45.9–36.6 mya) (Figure 3). The major clades and relationships of *Chamaecrista* that were previously recognised by Cota et al. (2025) were also strongly supported (pp = 1.0). The Adenophylum clade (*C. bucherae* the only extant species) was the first diverging lineage within *Chamaecrista*, dating back to 41.2 mya (95% HPD = 45.9–36.6 mya). The Apoucouita clade (12 spp.) diverged about 39.8 mya (95% HPD = 44.5–36.0 mya) from a major clade bringing together the remaining *Chamaecrista* lineages. The Absus clade (150 spp.) diverged from Chamaecrista + Baseophyllum + Otophyllum clade about 35.9 mya (95% HPD = 40.9–32.1 mya). The Chamaecrista clade (59 spp.) diverged from the Baseophyllum + Otophyllum clade about 29.4 mya (95% HPD = 32.0–25.3 mya). Finally, the Baseophyllum (8 spp.) and Otophyllum (*C. debilis* the only extant species) clades diverged from each other about 21 mya (95% HPD = 32.0–17.7 mya).

### 3.2 | Biogeographic Reconstruction

The DEC3 model was the best-fit (AIC = 1639.34) for the phylogenetic tree and geographic distribution of *Chamaecrista* species, exhibiting the highest conditional probabilities among the nine models evaluated (AICw = 77%) (Table 1). The DEC3 model had three times more support than the second-best model, the DEC2 model. Under the DEC3 model, the *campos rupestres* were available for colonisation before the Late Oligocene and Mid Miocene (28–10 Mya), preceding the expansion of other open-vegetation areas in the Neotropics. Dispersal rates were overall low (d = 0.029), whereas local extinction rates were high (*e* = 0.106) (Data S6).

Considering the DEC3 model (Figure 4), *Chamaecrista* diverged from its sister lineage most likely in the Amazon about 54.2 mya (Figure 4, diamond mark 1), where it later diversified about 41.2 mya. The Adenophyllum clade, currently restricted to Central America, likely originated in the Amazon about 41.2 mya (Figure 4, diamond mark 2). Similarly, the Apoucouita clade, currently in the Amazon and Atlantic Forest, likely originated in the Amazon between 39.8 and 35.3 mya (Figure 4, diamond mark 3). Within the Apocouita clade, a single ancestral lineage dispersed to the Atlantic Forest and became extinct in the Amazon between 35.3 and 26.2 mya, with a more recent lineage dispersing back to the Amazon after 24.7 mya. In contrast, the Absus + Chamaecrista + Baseophyllum + Otophyllum clade, mostly in neotropical open-vegetation areas, dispersed from the Amazon to the *campos rupestres* between 39.8 and 35.9 mya (Figure 4, diamond mark 4), later diversifying in two major clades. First, the Chamaecrista + Baseophyllum + Otophyllum clade likely originated in the *campos rupestres* between 35.9 and 29.4 mya (Figure 4, diamond mark 5). The Baseophyllum + Otophyllum clade remained and diversified in the *campos rupestres* (Figure 4, diamond mark 6), with more recent lineages dispersing to the Cerrado and Caatinga after 10 mya. The Chamaecrista clade originated in an underdetermined area between 29.4 and 18 mya (Figure 4, diamond mark 7), later dispersing to the *campos rupestres*, Cerrado, and Caatinga areas between 18 and 10 mya. Within the Chamaecrista clade, three ancestral lineages dispersed to extra-American areas between 8.6 and 2.5 mya, and one of them became extinct in the Neotropics and originated a major clade restricted to extra-American areas (Figure 4, diamond mark 8). The Absus clade likely originated in the *campos rupestres* between 35.9 and 16.0 mya (Figure 4, diamond mark 9). Within the Absus clade, a single ancestral lineage dispersed to the Cerrado between 14 and 12 mya. Later, ancestral lineages dispersed back to the *campos rupestres* after 6.5 mya, with further dispersal in the other direction after 2.5 mya.

## 4 | Discussion

By explicitly testing alternative models, our study elucidated how ancient climatic processes shaped the biogeographic history of *Chamaecrista* across the Neotropics. The best-fitting model supported a time-stratified scenario where the *campos rupestres* were colonised before the Late Oligocene–Mid Miocene (28–10 Mya), preceding the expansion of other open-vegetation areas in the Neotropics. Our results establish the *campos rupestres* as an ancient montane reservoir that supplied plant diversity to the surrounding lowlands. Furthermost, our study offers a nuanced understanding of how historical processes contributed to the assembly of species-rich biotas in open-vegetation areas in the Neotropics.

### 4.1 | Origins in Rainforests

Our results indicate that *Chamaecrista* originated and initially diversified in the Amazon during the Mid Eocene (~54 mya), aligning with a previous multi-taxon biogeographic analysis that recognises Amazonia as a cradle of neotropical biodiversity (Antonelli et al. 2018). Despite its origin and early diversification in the Amazon, *Chamaecrista* is considerably less diverse in this region today than in neotropical open-vegetation areas. This pattern may reflect distinct diversification dynamics associated with rainforest and open-vegetation areas, a hypothesis awaiting a formal test with diversification models. The Amazon accumulates great edaphic heterogeneity (Quesada et al. 2020), primarily due to orogenic processes related to the Andes uplift (Hoorn et al. 2010; Pérez-Escobar et al. 2022). This edaphic heterogeneity, particularly soil-phosphorus variation, predicts plant species distributions in the Amazon (Tuomisto et al. 1995, 2016), also influencing



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

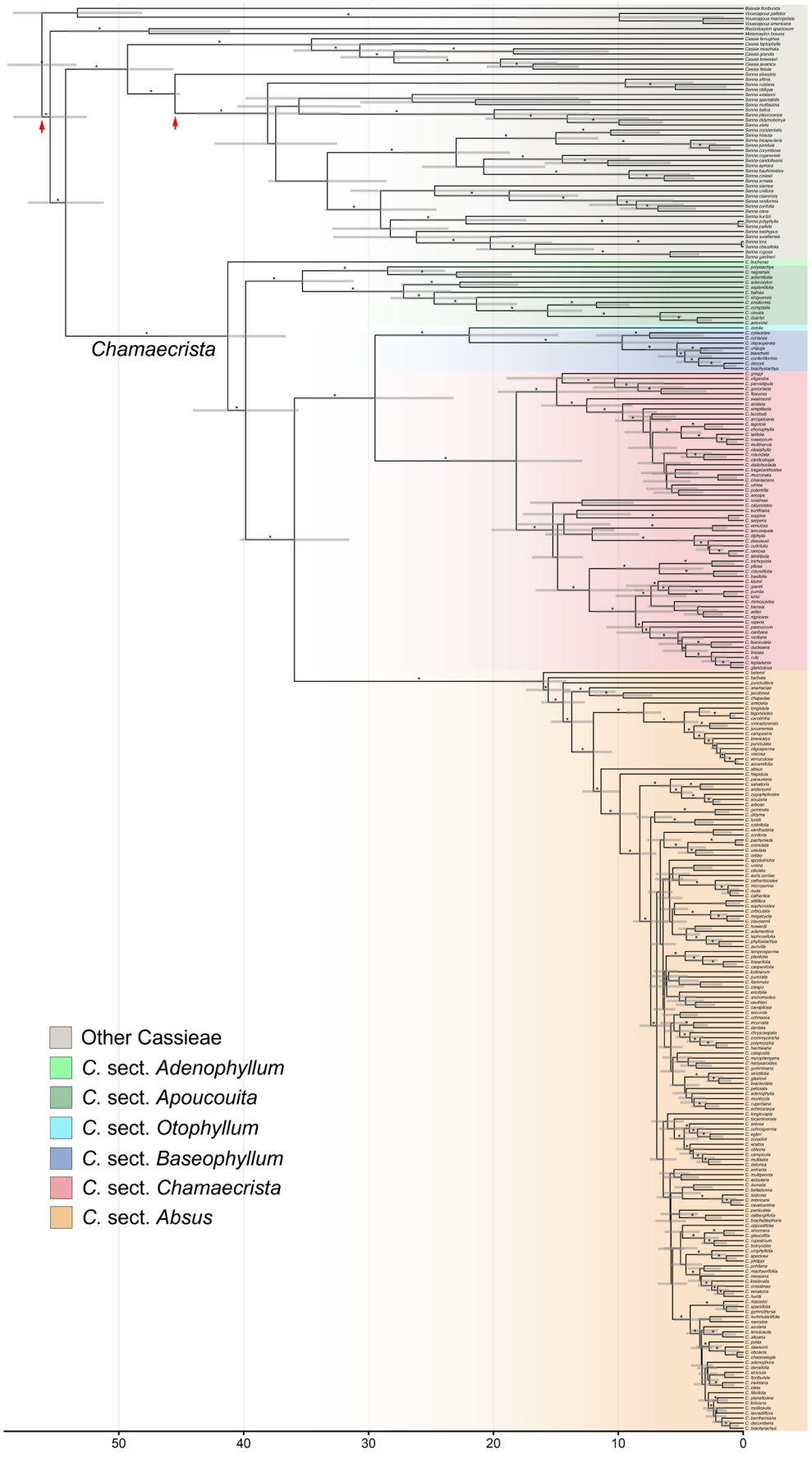



**FIGURE 3** | Legend on next page.



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**FIGURE 3** | Divergence time of neotropical *Chamaecrista* species as estimated from the BEAST analysis of the nuclear loci ITS and ETS and the plastidial loci *trnD-T*, *trnL-F*, and *trnK-matK*. Asterisks (*) over branches indicate posterior probabilities equal or greater than 0.95 based on 4.000 BEAST-generated trees. The node bars indicate the 95% HPD for age estimates. The scale below indicates time in millions of years. The red arrows indicate the calibration nodes.

**TABLE 1** | The nine biogeographic models fitted to the MCC phylogenetic tree and distribution of *Chamaecrista* species, followed by their respective biogeographic scenarios and goodness-of-fit. Biogeographic scenarios considered different times for the colonisation of the *campos rupestres* (CR) area. The best-fitting model is in bold.

| Model | Biogeographic scenario | LnL | AIC | AICw |
|---|---|---|---|---|
| DEC1 | CR and other areas equally available throughout time, dispersal unconstrained | −870.790 | 1745.58 | 0 |
| DEC2 | CR area available from the Late Oligocene—Mid Miocene (28–10 Mya) onwards, dispersal constrained among non-neighbour areas | −818.892 | 1641.78 | 22 |
| **DEC3** | **CR area available before the Late Oligocene—Mid Miocene (28–10 Mya), dispersal constrained among non-neighbour areas** | **−817.670** | **1639.34** | **77** |
| DIVALIKE1 | CR and other areas equally available throughout time, dispersal unconstrained | −969.974 | 1943.94 | 0 |
| DIVALIKE2 | CR area available from the Late Oligocene—Mid Miocene (28–10 Mya) onwards, dispersal constrained among non-neighbour areas | −912.061 | 1828.12 | 0 |
| DIVALIKE3 | CR area available before the Late Oligocene—Mid Miocene (28–10 Mya), dispersal constrained among non-neighbour areas | −910.104 | 1824.21 | 0 |
| BAYAREA1 | CR and other areas equally available throughout time, dispersal unconstrained | −869.442 | 1742.88 | 0 |
| BAYAREA2 | CR area available from the Late Oligocene—Mid Miocene (28–10 Mya) onwards, dispersal constrained among non-neighbour areas | −870.306 | 1744.61 | 0 |
| BAYAREA3 | CR area available before the Late Oligocene—Mid Miocene (28–10 Mya), dispersal constrained among non-neighbour areas | −866.887 | 1737.77 | 0 |

the assemblage of nitrogen-fixing symbionts for legumes (e.g., Pires et al. 2018; Rouws et al. 2024). Consequently, Amazonian soil types are inferred to promote plant species diversification by edaphic specialisation (e.g., Fine et al. 2005, 2014). Hence, we propose that edaphic specialisation likely promoted the early divergence of *Chamaecrista* species in the Amazon, later facilitating species diversification within the Apoucouita clade.

An ancestral *Chamaecrista* lineage dispersed from the Amazon to the Atlantic Forest during the Oligocene and Early Miocene (35–26 mya), supporting an ancestral connection between these two rainforest dominions. The spatial connection between these two rainforest dominions was likely greater before the Late Oligocene–Mid Miocene (28–10 Mya), when climatic shifts presumably promoted the formation of the open diagonal (Gottsberger and Silberbauer-Gottsberger 1988). The open diagonal likely acted as a reproductive barrier for plant populations in rainforests, promoting vicariance between the Amazon and Atlantic Forest dominions (Guillory et al. 2024). Supporting this scenario, an ancestral *Chamaecrista* lineage from the Amazon dispersed and became endemic to the Atlantic Forest upon the onset of open-vegetation areas, while its sister lineage remained endemic to the Amazon. Still, the ancestral lineage of *Chamaecrista bahiae* (H.S.Irwin) H.S.Irwin & Barneby dispersed from the Atlantic Forest back to the Amazon afterwards, indicating an enduring connection between the Amazon and Atlantic Forest even with the open diagonal. An interdigitated matrix of gallery forests that withstand dry climates may have enabled this historical connection (Ledo and Colli 2017). Similarly, many extant species have a disjunct distribution between the Amazon and Atlantic Forest, as extensively documented, for example, in birds (Batalha-Filho et al. 2013), mammals (Vivo 1997), and vascular plants (Andrade-Lima 1982; Prance 1982; Martini et al. 2007; Bacci et al. 2022; Nicholls et al. 2025). In this context, the varying spatial connectivity between the Amazon and Atlantic Forest likely facilitated both ancestral vicariance and more recent dispersal within the Apocouita clade of *Chamaecrista*.

### 4.2 | Recent Diversification in Open Vegetations

While the Amazon rainforest was the cradle, open-vegetation areas fostered most *Chamaecrista* diversification. An ancestral *Chamecrista* lineage first colonised the *campos rupestres* before the Late Oligocene—Mid Miocene (28–10 Mya), preceding the expansion of lowland open-vegetation in the Neotropics. The major clades within *Chamaecrista* primarily diverged in the *campos rupestres* before the above mentioned period. However, most diversification in *Chamaecrista* occurred from the Late Miocene to the Quaternary. Similarly, multiple plant lineages endemic to the *campos rupestres* have been inferred to primarily derive from recent bursts of species diversification (~5 Mya) (Vasconcelos et al. 2020). The increase in situ speciation in the *campos rupestres* has been explained by an expansion-contraction dynamic



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

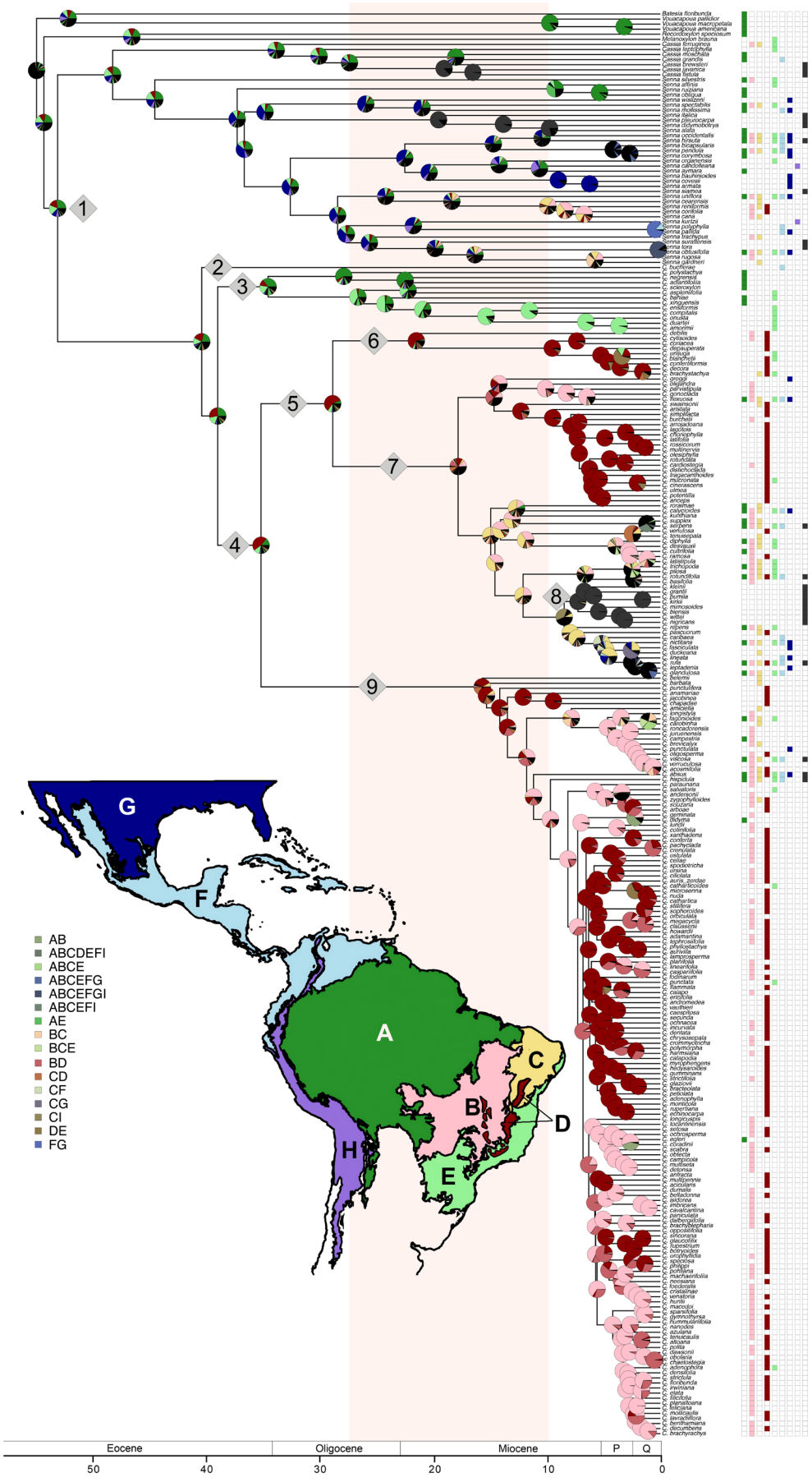



**FIGURE 4** | Legend on next page.



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**FIGURE 4** | Biogeographic reconstruction of neotropical *Chamaecrista* species based on the DEC3 model. Coloured squares at the tips indicate the known geographic states of extant species. Pie charts at nodes indicate the marginal likelihood of geographic states for the MRCA (most recent common ancestor). The map depicts the geographic areas in our biogeographic analysis: (A) Amazon; (B) Cerrado; (C) Caatinga; (D) *Campos rupestres*; (E) Atlantic Forest; (F) Central America; (G) North America; (H) Andes. The scale below indicates time in millions of years and the corresponding geological periods. The vertical light strip highlights the second time frame of our biogeographic analysis, the Late Oligocene—Mid Miocene (28–10 Mya). Diamond marks indicate historical processes narrated in the main text: (1) the divergence between *Chamaecrista* and its sister taxa in the Amazon; (2) the stem lineage marking the origin of the Adenophyllum clade in the Amazon; (3) the origin of the Apocouita clade in the Amazon; (4) the dispersal of the ancestral lineage of the Absus + Chamaecrista + Baseophyllum + Otophyllum to the *campos rupestres*; (5) the origin of the Chamaecrista + Baseophyllum + Otophyllum clade in the *campos rupestres*; (6) the origin of Baseophyllum + Otophyllum clade in the *campos rupestres*; (7) the origin of the Chamaecrista clade in an undetermined area; (8) the dispersal of an ancestral lineage within the Chamaecrista clade to extra-American areas; and (9) the origin of the Absus clade in the *campos rupestres*.

of plant populations during the glacial-interglacial cycles of the Plio-Pleistocene periods (Vasconcelos et al. 2020). However, montane areas of the Cerrado dominion, including some *campos rupestres*, were associated with greater climatic stability between glacial and interglacial ages than nearby lowland areas (Vidal et al. 2019). This greater climatic stability had a minor impact over the geographic distribution of woody plant species in montane Cerrado settings and some *campos rupestres* (Silva et al. 2024), perhaps precluding long-term geographic isolation and speciation. Hence, glacial–interglacial cycles might have imposed a mild expansion-contraction dynamic in the *campos rupestres*, having a negligible effect over the diversification of *Chamaecrista*.

Alternatively, fire regimes could have also contributed to the recent diversification of *Chamaecrista* in the *campos rupestres*. Fires became presumably more common in neotropical savanna-like open-vegetation during the Pliocene (Pennington et al. 2006), likely fueled by the expansion and accumulation of C4 grasses during the Miocene (Edwards et al. 2010). These fires imposed novel evolutionary regimes upon plant lineages in savannas of the Cerrado, prompting species diversification (Simon et al. 2009). Likewise, the *campos rupestres* could have fostered an increased plant species diversification after the establishment of natural fire regimes (Simon and Pennington 2012). For instance, in a specific clade of *Chamaecrista*, species evolved underground storage organs when under greater fire frequencies, indicating a selective pressure of fire regimes upon plant species (Prochazka et al. 2024). Hence, fire-driven selective pressures could have also promoted the diversification of *Chamaecrista*, a hypothesis awaiting further testing for its *campos rupestres* lineages.

### 4.3 | The *Campos Rupestres* as a Montane Reservoir of Plant Diversity

Our analyses support the *campos rupestres* as an ancient montane area that fostered lineage persistence and diversification. The *campos rupestres* were initially colonised during the Late Eoceno, preceding the expansion of lowland open-vegetation areas in the Neotropics. After the initial colonisation, the *campos rupestres* fostered most *Chamaecrista* diversification, originating two major clades. Later, multiple ancestral lineages dispersed from the *campos rupestres* to lowland open-vegetation areas. These results support the *campos rupestres* as montane reservoirs of plant diversity (Perrigo et al. 2020).

Unlike most lowland savannas, the *campos rupestres* are associated with ancient and weathered montane substrates that date back to the Paleoproterozoic (Alkmim 1998; Pedreira and de Waele 2008). Over geological timescales, these uplands have developed a complex mosaic of rocky outcrops and edaphically divergent habitats, promoting strong environmental filtering, spatial isolation, and ecological specialisation (e.g., le Stradic et al. 2015). This ecological filtering likely restricted the species' capacity to establish beyond these uplands, contributing to the exceptional levels of local endemism observed today (Rapini et al. 2021). These conditions are also well-established drivers of endemism and speciation in montane ecosystems more broadly (Muellner-Riehl et al. 2019; Nery et al. 2023), and they likely influenced the evolutionary trajectories of multiple *Chamaecrista* lineages in the *campos rupestres*. Supporting this perspective, phosphorus availability (or the lack of it) is correlated to distinct nutritional strategies among plant communities in the *campos rupestres* (Alcantara et al. 2015; Oliveira et al. 2015), also distinguishing these plant communities from those in the savanna of the Cerrado dominion (Abrahão et al. 2019). Closely juxtapositioned areas of *campos rupestres* can be very different in floristic composition, forming distinct areas of plant endemism (Echternacht et al. 2011), which most likely reflects this edaphic specialisation. Edaphic specialisation also may have evolved among *Chamaecrista* lineages as their plant life and habitat strategies have been inferred to selectively evolve in response to soil conditions (Prochazka et al. 2024). For instance, some distantly-related species evolved nitrogen-limited niches when colonising the *campos rupestres* (Prochazka et al. 2024), indicating selective pressures favouring edaphic specialisation. These nitrogen-limited niches can also reflect the symbiosis between *Chamaecrista* species and nitrogen-fixing bacteria (Casaes et al. 2024), which would be most favourable under nitrogen-limiting conditions (Kobayashi et al. 2021; Souza et al. 2024). If so, the symbiosis with nitrogen-fixing bacteria could have facilitated the colonisation of the *campos rupestres*, where soil nitrogen is often limiting (Silveira et al. 2016). Therefore, edaphic specialisation, likely facilitated by mutualistic associations with nitrogen-fixing bacteria, may have enabled ancestral *Chamaecrista* lineages to establish on nutrient-poor substrates of the *campos rupestres*. By successfully colonising one of the many edaphically distinct habitats in these ancient montane systems, they could have given rise to locally endemic lineages well before the expansion of colonisable lowland open-vegetation areas.

After the initial colonisation, the environmental heterogeneity of the *campos rupestres* likely favoured a long-term persistence



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of endemic plant lineages. The environmental heterogeneity of montane landscapes presumably allows plant species to rapidly track their suitable habitats during climatic oscillations, promoting long-term persistence over extinction (Perrigo et al. 2020). Consequently, mountain regions with greater elevation ranges overall harbour greater plant species diversity (Muellner-Riehl et al. 2019). In the *campos rupestres*, plant communities are marked by distinct strategies over the elevation gradient (Mota et al. 2018), reflecting the environmental heterogeneity of montane landscapes (Silveira et al. 2016). Such an environmental heterogeneity would have allowed endemic plant species to persist in the *campos rupestres* during past climatic shifts (Rapini et al. 2021), including those promoting the formation of the open diagonal. This long-term persistence would have allowed a greater accumulation of endemic plant species in the *campos rupestres* than in surrounding lowland areas (Vidal et al. 2019). This scenario would translate into lower extinction rates for *Chamaecrista* species endemic to the *campos rupestres*, a hypothesis awaiting a formal test. Nonetheless, other plant lineages had experienced faster diversification upon colonising the *campos rupestres* (e.g., Magri et al. 2025), possibly due to a lower extinction in these montane landscapes.

With the formation of the open diagonal, the campos rupestres likely acted as a major source area for savanna vegetation within the Cerrado dominion. Four to five ancestral *Chamaecrista* lineages dispersed from the *campos rupestres* to the Cerrado from the Mid Miocene onwards, while three to four lineages dispersed in the opposite direction from the Late Miocene onwards. This pattern highlights a great floristic exchange between the *campos rupestres* and the savannas of Cerrado, leading to their known strong floristic links (Neves et al. 2018). Such a floristic exchange was likely facilitated by the similar abiotic conditions in the *campos rupestres* and the Cerrado savannas (Silveira et al. 2016). First, both vegetation types have nutrient-poor soils (e.g., Abrahão et al. 2019). Second, both vegetation types are subject to natural fires, yet under lower frequency in the *campos rupestres*, where rock outcrops would hinder fire ignition and spread (Conceição et al. 2016). Supporting this perspective, both soil conditions and fire frequency had impacted the evolution of life forms of *Chamaecrista* species that can occur in both the *campos rupestres* and Cerrado (Prochazka et al. 2024). Given its more recent formation, the Cerrado dominion has been considered a recipient area for plant lineages originating in other surrounding areas (Simon and Pennington 2012), yet the *campos rupestres* have not been considered a donor area until now. Here we showed that ancestral *Chamaecrista* species dispersing from the *campos rupestres* likely had traits that facilitated the colonisation of lowland savannas, translating into multiple dispersals between these two similar areas.

In contrast, the *campos rupestres* likely acted as a minor source area for the Caatinga. Two ancestral *Chamecrista* lineages dispersed from the *campos rupestres* to the Caatinga from the Late Miocene onwards, without subsequent dispersals into the opposite direction. This limited floristic exchange is likely due to the contrasting ecology between the two areas. Unlike other neighbouring open-diagonal formations, the Caatinga seasonally dry forests experience a more arid climate and are less influenced by fire regimes (Werneck 2011). These constraints have shaped the Caatinga plant species composition as strongly filtered by aridity (Silva and Souza 2018). Consequently, aridity likely favoured colonisation into the Caatinga by plant lineages from other Neotropical seasonally dry forests over those from adjacent ecologically contrasting areas (Pennington et al. 2000). The ancestral *Chamaecrista* lineages that colonise the Caatinga diversified there during the Late Miocene. These were predominantly annual life forms, which employ a rapid life cycle to germinate, grow, and reproduce by exploiting seasonal rains before senescence (Prochazka et al. 2024), thereby an ecologically successful strategy to cope with the Caatinga's severe arid period. The spread of such short-lived, fast-growing plants with high nutrient demands across the Caatinga was likely facilitated by the shallow, richer soils of the crystalline basement (Santos et al. 2012) or through associations with mutualistic root symbionts. This pattern supports scenarios of in situ speciation in the Caatinga (Fernandes et al. 2022), driven by edaphic specialisation, which is a key factor influencing the region's floristic affinities and biogeographic regionalization (Moro et al. 2016, 2024). Therefore, the ancestral *Chamaecrista* species that dispersed from the *campos rupestres* likely had to evolve novel traits to colonise the Caatinga, explaining the rarity of dispersal events between these two areas.

Despite these different dynamics, the *campos rupestres* fostered ancestral *Chamaecrista* lineages that later dispersed to lowland open-vegetation areas, resembling mountain regions worldwide (Perrigo et al. 2020). Mountain regions are often more biodiverse than neighbouring lowland areas (Perrigo et al. 2020). Montane biotas are primarily formed by endemic species that presumably derive from specialisation and in situ speciation over topographically complex landscapes (Rahbek et al. 2019). For instance, the environmental heterogeneity of montane landscapes favoured the evolution of plant endemism in the Atlantic Forest (e.g., Nery et al. 2023). Consequently, the Atlantic Forest accumulates a great diversity of small-ranged plant species (Leão et al. 2020), becoming a source area for neighbouring dominion (e.g., Santos et al. 2017). Similarly, our findings indicate that the *campos rupestres* have served as a montane reservoir of plant diversity for neotropical lowland open-vegetation areas, challenging previous assumptions that tropical montane vegetation is secondary or derivative forms of lowland vegetation.

### 4.4 | Extra-American Dispersals

Despite the predominant distribution in the Neotropics, few *Chamaecrista* species ($n=9$) occur in extra-American areas, mostly Africa, with a single clade being exclusive of these areas. This extra-American clade ($n=7$) originated from an ancestral species dispersing from the Caatinga during the Late Miocene (8–7 mya), later becoming extinct in the Neotropics. Other two ancestral species that were widespread in the Neotropics dispersed to extra-American areas during the Pliocene and Quaternary (5–2.5 mya). These trans-oceanic dispersals might reflect multiple processes (De Queiroz 2005). First, seeds might have been directly transported between continents by sea currents and strong winds (Renner 2004). For instance, the North Equatorial Counter Current (NECC) flows from northern South America to West Africa, potentially promoting rafting between these two continents (Renner 2004). Similarly, westerly wind jets can occur over the South American coast and reach West Africa, potentially dragging small seeds (Renner 2004). Second, ancestral species



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

might have progressively colonised islands until reaching another continent, i.e., 'stepping stones' (Saura et al. 2014). For instance, the islands could have acted as stepping stones between the north-eastern coast of Brazil and the west coast of Africa. However, our current sampling of extra-American *Chamaecrista* species is too small, precluding a deeper understanding of these trans-oceanic dispersals. To address this issue, future studies should incorporate a broader sampling of *Chamaecrista* from Africa, Asia, and Oceania. Importantly, our results show that the extra-American lineages trace back to neotropical open-vegetation areas rather than rainforests. Hence, early dispersing *Chamaecrista* lineages likely had traits that facilitated the colonisation of drought-prone habitats in extra-American areas.

## 5 | Conclusion

Here, we investigated the biogeographic history of the species-rich legume genus *Chamaecrista*, as a model for understanding how plant diversity has accumulated in neotropical open-vegetation areas in the Neotropics. The genus diverged from its sister lineage in the Amazon during the Mid Eocene. Later, *Chamaecrista* species colonised the *campos rupestres* during the Late Eocene, before the expansion of lowland open-vegetation areas in the Neotropics. In the *campos rupestres*, *Chamaecrista* species diversified and later dispersed to the Cerrado and Caatinga from the Miocene onwards. Our findings underscore the *campos rupestres* as montane reservoirs of plant lineages for neotropical lowland open-vegetation areas. Moreover, our findings support recent regionalization proposals that recognise the *campos rupestres* as an independent biogeographic unit, given its distinct evolutionary history. In this context, the *campos rupestres* emerge as an important source of diversification for plant lineages that later expanded into lowland open-vegetation areas.

### Author Contributions

J.G.R.: conceptualisation, methodology, formal analysis, figures, writing – original draft, data curation. E.K.N.: methodology, formal analysis, figures, writing – original draft. A.N.: methodology, writing – review and editing. M.M.T.C.: methodology. A.Z.: methodology, formal analysis, writing – review and editing. S.A.: writing – review and editing. G.P.L.: methodology, writing – original draft. D.C.: writing – review and editing. L.P.Q.: conceptualisation, writing – review and editing. J.R.P.: conceptualisation, writing – review and editing.


### Acknowledgements

This research originated during the first author's doctoral studies completed in 2014, which were fully supported by the São Paulo Research Foundation (FAPESP; grants 2009/16434–4 and 2013/09374–0). Additional support was provided by the Research Support Foundation of the State of Bahia (FAPESB; grant JCB0030/2016). The first author (JGR) is currently supported by a FAPESB postdoctoral fellowship (grant BOL1132/2025). A.N. is supported by grants from the São Paulo Research Foundation (FAPESP; grant 2019/19544-7), the National Council for Scientific and Technological Development (CNPq; grant 312389/2023-0), and the Serrapilheira Institute (Public Call No. 7–2023). E.K.N. acknowledges support from FAPESP (grant 2023/17719-0). D.C. is supported by CNPq (grant 314187/2021-9) and FAPERJ (grant E-26/200.153/2023). The Article Processing Charge for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614).

### Funding

This work was supported by Fundação de Amparo à Pesquisa do Estado de São Paulo, 2009/16434-4, 2013/09374-0, 2019/19544-7, 2023/17719-0. Fundação de Amparo à Pesquisa do Estado da Bahia, JCB0030/2016, BOL1132/2025. Conselho Nacional de Desenvolvimento Científico e Tecnológico, 312389/2023-0, 314187/2021-9. Serrapilheira Institute (Public Call No. 7-2023). The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) (ROR identifier:00x0ma614). For the purposes of open access, the authors have applied a Creative Commons CC BY license to any accepted version of the article.


### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

All data obtained in this study are available in the Data S1. The scripts and data used are available at https://github.com/eknery/Chamaecrista_biogeography and https://figshare.com/projects/From_rainforests_to_open-vegetation_The_biogeographic_History_of_i_Chamaecrista_i_Leguminosae_in_the_Neotropics/252935. Any additional information can be directly obtained by contacting the first author of this study.

### References


Abrahão, A., P. B. Costa, H. Lambers, et al. 2019. "Soil Types Select for Plants With Matching Nutrient-Acquisition and -Use Traits in Hyperdiverse and Severely Nutrient-Impoverished Campo Rupestre and Cerrado in Central Brazil." *Journal of Ecology* 107: 1302–1316. https://doi.org/10.1111/1365-2745.13111.

Akaike, H. 1974. "A New Look at the Statistical Model Identification." *IEEE Transactions on Automatic Control* 19: 716–723. https://doi.org/10.1109/TAC.1974.1100705.

Alcantara, S., R. Mello-Silva, G. S. Teodoro, K. Drequeceler, D. D. Ackerly, and R. S. Oliveira. 2015. "Carbon Assimilation and Habitat Segregation in Resurrection Plants: A Comparison Between Desiccation-and Non-Desiccation-Tolerant Species of Neotropical Velloziaceae (Pandanales)." *Functional Ecology* 29: 1499–1512. https://doi.org/10.1111/1365-2435.12462.

Alencar, C. L. S., A. Nogueira, R. E. Vicente, and I. A. C. Coutinho. 2013. "Plant Species With Larger Extrafloral Nectaries Produce Better Quality Nectar When Needed and Interact With the Best Ant Partners." *Journal of Experimental Botany* 74, no. 15: 4613–4627. https://doi.org/10.1093/jxb/erad160.

Alkmim, F. F. 1998. "Transamazonian Orogeny in the Southern São Francisco Craton Region, Minas Gerais, Brazil: Evidence for Paleoproterozoic Collision and Collapse in the Quadrilátero Ferrífero." *Precambrian Research* 90: 29–58. https://doi.org/10.1016/s0301-9268(98)00032-1.

Alves, R. J. V., N. G. Silva, J. A. Oliveira, and D. Medeiros. 2014. "Circumscribing Campo Rupestre – Megadiverse Brazilian Rocky Montane Savanas." *Brazilian Journal of Biology* 74: 355–362. https://doi.org/10.1590/1519-6984.23212.

Andrade-Lima, D. 1982. "Present-Day Forest Refuges in Northeastern Brazil." In *Biological Diversification in the Tropics*, edited by G. T. Prance, 247–251. Columbia University Press.

Antonelli, A. 2022. "The Rise and Fall of Neotropical Biodiversity." *Botanical Journal of the Linnean Society* 199: 8–25. https://doi.org/10.1093/botlinnean/boab061.

Antonelli, A., J. A. A. Nylander, C. Persson, and I. Sanmartín. 2009. "Tracing the Impact of the Andean Uplift on Neotropical Plant





13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Evolution." *Proceedings of the National Academy of Sciences* 106: 9749–9754. https://doi.org/10.1073/pnas.0811421106.

Antonelli, A., and I. Sanmartín. 2011. "Why Are There So Many Plant Species in the Neotropics?" *Taxon* 60: 403–414. https://doi.org/10.1002/tax.602010.

Antonelli, A., A. Zizka, F. A. Carvalho, and F. L. Condamine. 2018. "Amazonia Is the Primary Source of Neotropical Biodiversity." *Proceedings of the National Academy of Sciences* 115: 6034–6039. https://doi.org/10.1073/pnas.1713819115.

Bacci, L. F., M. Reginato, T. Bochorny, F. A. Michelangeli, A. M. Amorim, and R. Goldenberg. 2022. "Biogeographic Breaks in the Atlantic Forest: Evidence for Oligocene/Miocene Diversification in Bertolonia (Melastomataceae)." *Botanical Journal of the Linnean Society* 199: 128–143. https://doi.org/10.1093/botlinnean/boab099.

Bachman, S. P., M. J. M. Brown, T. C. C. Leão, E. Nic Lughadha, and B. E. Walker. 2024. "Extinction Risk Predictions for the World's Flowering Plants to Support Their Conservation." *New Phytologist* 242: 797–808. https://doi.org/10.1111/nph.19592.

Barbosa, B. C., T. D. D. Lima, G. V. Mota, and A. Nogueira. 2023. "The Role of Intraspecific Variation in Bumblebee Body Size and Behavior on Buzz Pollination of a Tropical Legume Species." *American Journal of Botany* 110, no. 11: e16236. https://doi.org/10.1002/ajb2.16236.

Barros, M. J. F., G. A. Silva-Arias, A. L. A. Segatto, et al. 2020. "Phylogenetic Niche Conservatism and Plant Diversification in South American Subtropical Grasslands Along Multiple Climatic Dimensions." *Genetics and Molecular Biology* 43: e20180291. https://doi.org/10.1590/1678-4685-GMB-2018-0291.

Batalha-Filho, H., J. Fjeldså, P. H. Fabre, and C. Y. Miyaki. 2013. "Connections Between the Atlantic and the Amazonian Forest Avifaunas Represent Distinct Historical Events." *Journal of Ornithology* 154: 41–50. https://doi.org/10.1007/s10336-012-0866-7.

Bouckaert, R., T. G. Vaughan, J. Barido-Sottani, et al. 2019. "BEAST 2.5: An Advanced Software Platform for Bayesian Evolutionary Analysis." *PLoS Computational Biology* 15: e1006650. https://doi.org/10.1371/journal.pcbi.1006650.

Bruneau, A., M. Mercure, G. P. Lewis, and P. S. Herendeen. 2008. "Phylogenetic Patterns and Diversification in the Caesalpinioid Legumes." *Botany* 86: 697–718. https://doi.org/10.1139/B08-058.

Bünger, M. O., F. F. Mazine, F. Forest, M. L. Bueno, J. R. Stehmann, and E. J. Lucas. 2016. "The Evolutionary History of *Eugenia* Sect. *Phyllocalyx* (Myrtaceae) Corroborates Historically Stable Areas in the Southern Atlantic Forests." *Annals of Botany* 118: 1209–1223. https://doi.org/10.1093/aob/mcw209.

Burnham, K. P., and D. R. Anderson. 2002. *Model Selection and Inference: A Practical Information-Theoretic Approach*. 2nd ed. Springer-Verlag. https://doi.org/10.1007/b97636.

Casaes, P. A., J. M. F. Santos, V. C. Silva, et al. 2024. "The Radiation of Nodulated *Chamaecrista* Species From the Rainforest Into More Diverse Habitats Has Been Accompanied by a Reduction in Growth Form and a Shift From Fixation Threads to Symbiosomes." *Journal of Experimental Botany* 75, no. 11: 3643–3662. https://doi.org/10.1093/jxb/erae134.

Chamberlain, S., V. Barve, D. Mcglinn, et al. 2025. "rgbif: Interface to the Global Biodiversity Information Facility API." R package version 3.8.1. https://CRAN.R-project.org/package=rgbif.

Chinarelli, H. D., A. Nogueira, and L. C. Leal. 2022. "Extrafloral Nectar Production Induced by Simulated Herbivory Does Not Improve Ant Bodyguard Attendance and Ultimately Plant Defence." *Biological Journal of the Linnean Society* 135, no. 3: 429–446. https://doi.org/10.1093/biolinnean/blab159.

Colli-Silva, M., T. N. C. Vasconcelos, and J. R. Pirani. 2019. "Outstanding Plant Endemism Levels Strongly Support the Recognition of *Campo Rupestre* Provinces in Mountaintops of Eastern South America." *Journal of Biogeography* 46: 1723–1733. https://doi.org/10.1111/jbi.13585.

Conceição, A. A., A. Rapini, F. F. Carmo, et al. 2016. "Rupestrian Grassland Vegetation, Diversity, and Origin." In *Ecology and Conservation of Mountaintop Grasslands in Brazil*, edited by G. W. Fernandes, 105–127. Springer International Publishing. https://doi.org/10.1007/978-3-319-29808-5_6.

Conceição, A. S., L. P. Queiroz, G. P. Lewis, et al. 2009. "Phylogeny of *Chamaecrista* Moench (Leguminosae-Caesalpinioideae) Based on Nuclear and Chloroplast DNA Regions." *Taxon* 58: 1168–1180. https://doi.org/10.1002/tax.584010.

Cota, M. M. T., J. G. Rando, D. Cardoso, A. S. Conceição, C. Snak, and L. P. Queiroz. 2025. "Circumscribing *Chamaecrista* Section *Baseophyllum* (Caesalpinioideae, Leguminosae) Based on Molecular Data and Morphology." *Botanical Journal of the Linnean Society* 209: 146–163. https://doi.org/10.1093/botlinnean/boaf007.

Crisp, M. D., N. B. Hardy, and L. G. Cook. 2014. "Clock Model Makes a Large Difference to Age Estimates of Long-Stemmed Clades With no Internal Calibration: A Test Using Australian Grasstrees." *BMC Evolutionary Biology* 14: 263. https://doi.org/10.1186/s12862-014-0263-3.

De Queiroz, A. 2005. "The Resurrection of Oceanic Dispersal in Historical Biogeography." *Trends in Ecology & Evolution* 20: 68–73. https://doi.org/10.1016/j.tree.2004.11.006.

Dutra, V. F., M. F. Vieira, F. C. P. Garcia, and H. C. Lima. 2009. "Fenologia Reprodutiva, Síndromes de Polinização e Dispersão em Espécies de Leguminosae dos Campo Rupestre do Parque Estadual do Itacolomi, Minas Gerais, Brasil." *Rodriguésia* 60: 371–387. https://doi.org/10.1590/2175-7860200960210.

Echternacht, L., M. Trovó, C. T. Oliveira, and J. R. Pirani. 2011. "Areas of Endemism in the Espinhaço Range in Minas Gerais, Brazil." *Flora* 206: 782–791. https://doi.org/10.1016/j.flora.2011.04.003.

Edler, D., T. Guedes, A. Zizka, M. Rosvall, and A. Antonelli. 2016. "Infomap Bioregions: Interactive Mapping of Biogeographical Regions From Species Distributions." *Systematic Biology* 66: 197–204. https://doi.org/10.1093/sysbio/syw087.

Edwards, E. J., C. P. Osborne, C. A. E. Strömberg, et al. 2010. "The Origins of C4 Grasslands: Integrating Evolutionary and Ecosystem Science." *Science* 328: 587–591. https://doi.org/10.1126/science.1177216.

Fenster, C. B. 1991. "Gene Flow in *Chamaecrista fasciculata* (Leguminosae): I. Gene Dispersal." *Evolution* 45: 398–409. https://doi.org/10.1111/j.1558-5646.1991.tb04413.x.

Fenster, C. B., X. Vekemans, and O. J. Hardy. 2003. "Quantifying Gene Flow From Spatial Genetic Structure Data in a Metapopulation of *Chamaecrista fasciculata* (Leguminosae)." *Evolution* 57: 995–1007. https://doi.org/10.1111/j.0014-3820.2003.tb00311.x.

Fernandes, M. F., D. Cardoso, R. T. Pennington, and L. P. Queiroz. 2022. "The Origins and Historical Assembly of the Brazilian Caatinga Seasonally Dry Tropical Forests." *Frontiers in Ecology and Evolution* 10: 723286. https://doi.org/10.3389/fevo.2022.723286.

Fine, P. V. A., D. C. Daly, G. V. Muñoz, I. Mesones, and K. M. Cameron. 2005. "The Contribution of Edaphic Heterogeneity to the Evolution and Diversity of *Burseraceae* Trees in the Western Amazon." *Evolution* 59: 1464. https://doi.org/10.1111/j.0014-3820.2005.tb01796.x.

Fine, P. V. A., F. Zapata, and D. C. Daly. 2014. "Investigating Processes of Neotropical Rain Forest Tree Diversification by Examining the Evolution and Historical Biogeography of the Protieae (Burseraceae)." *Evolution* 68: 1988–2004. https://doi.org/10.1111/evo.12414.

Flora e Funga do Brasil. 2025. "Jardim Botânico do Rio de Janeiro." https://floradobrasil.jbrj.gov.br. Retrieved May 4, 2025.



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Fonseca, C. R., and E. M. Venticinque. 2018. "Biodiversity Conservation Gaps in Brazil: A Role for Systematic Conservation Planning." *Perspectives in Ecology and Conservation* 16: 61–67. https://doi.org/10.1016/j.pecon.2018.03.001.

Gadagkar, S. R., M. S. Rosenberg, and S. Kumar. 2005. "Inferring Species Phylogenies From Multiple Genes: Concatenated Sequence Tree Versus Consensus Gene Tree." *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution* 304B, no. 1: 64–74. https://doi.org/10.1002/jez.b.21026.

Gottsberger, G., and I. Silberbauer-Gottsberger. 1988. "Evolution of Flower Structure and Pollination in Neotropical Cassiinae (Caesalpiniaceae) Species." *Phyton* 28, no. 2: 293–320.

Guillory, W. X., F. De Medeiros Magalhães, F. E. A. Coelho, et al. 2024. "Geoclimatic Drivers of Diversification in the Largest Arid and Semi-Arid Environment of the Neotropics: Perspectives From Phylogeography." *Molecular Ecology* 33, no. 14: e17431. https://doi.org/10.1111/mec.17431.

Herendeen, P. S., W. L. Crepet, and D. L. Dilcher. 1992. "The Fossil History of the Leguminosae: Phylogenetic and Biogeographic Implications." In *Advances in Legume Systematics, Part 4, the Fossil Record*, 303–316. Royal Botanic Gardens, Kew.

Hoorn, C., F. P. Wesselingh, H. ter Steege, et al. 2010. "Amazonia Through Time: Andean Uplift, Climate Change, Landscape Evolution, and Biodiversity." *Science* 330: 927–931. https://doi.org/10.1126/science.11945.

Hortal, J., F. de Bello, J. A. F. Diniz-Filho, T. M. Lewinsohn, J. M. Lobo, and R. J. Ladle. 2015. "Seven Shortfalls That Beset Large-Scale Knowledge of Biodiversity." *Annual Review of Ecology, Evolution, and Systematics* 46: 523–549. https://doi.org/10.1146/annurev-ecolsys-112414-054400.

Hughes, C. E., R. T. Pennington, and A. Antonelli. 2013. "Neotropical Plant Evolution: Assembling the Big Picture." *Botanical Journal of the Linnean Society* 171: 1–18. https://doi.org/10.1111/boj.12006.

IBGE, Instituto Brasileiro de Geografia e Estatística. 2025. "Domínios e regiões naturais terrestres dos biomas do Brasil: Uma proposta de regionalização natural." https://biblioteca.ibge.gov.br/index.php/biblioteca-catalogo?view=detalhes&id=2102232.

Jaramillo, C. 2023. "The Evolution of Extant South American Tropical Biomes." *New Phytologist* 239: 477–493. https://doi.org/10.1111/nph.18931.

Katoh, K., and D. M. Standley. 2013. "MAFFT Multiple Sequence Alignment Software Version 7: Improvements in Performance and Usability." *Molecular Biology and Evolution* 30: 772–780. https://doi.org/10.1093/molbev/mst010.

Kobayashi, R., S. Yamaguchi, and Y. Iwasa. 2021. "Optimal Control of Root Nodulation – Prediction of Life History Theory of a Mutualistic System." *Journal of Theoretical Biology* 510: 110544. https://doi.org/10.1016/j.jtbi.2020.110544.

le Stradic, S., E. Buisson, and G. W. Fernandes. 2015. "Vegetation Composition and Structure of Some Neotropical Mountain Grasslands in Brazil." *Journal of Mountain Science* 12, no. 4: 864–877. https://doi.org/10.1007/s11629-013-2866-3.

Leão, T. C. C., E. N. Lughadha, and P. B. Reich. 2020. "Evolutionary Patterns in the Geographic Range Size of Atlantic Forest Plants." *Ecography* 43, no. 10: 1510–1520. https://doi.org/10.1111/ecog.05160.

Ledo, R. M. D., and G. R. Colli. 2017. "The Historical Connections Between the Amazon and the Atlantic Forest Revisited." *Journal of Biogeography* 44: 2551–2563. https://doi.org/10.1111/jbi.13049.

Lee, T. D. 1984. "Effects of Seed Number Per Fruit on Seed Dispersal in *Cassia fasciculata* (Caesalpiniaceae)." *International Journal of Plant Sciences* 145, no. 1: 136–139. https://doi.org/10.1086/337437.

Leigh, E. G., A. O'Dea, and G. J. Vermeij. 2013. "Historical Biogeography of the Isthmus of Panama." *Biological Reviews* 89: 148–172. https://doi.org/10.1111/brv.12048.

LPWG [Legume Phylogeny Working Group]. 2017. "A New Subfamily Classification of the Leguminosae Based on a Taxonomically Comprehensive Phylogeny." *Taxon* 66: 44–77. https://doi.org/10.12705/661.3.

Magri, R. A., F. Luebert, A. Cabral, et al. 2025. "Historical Biogeography of Vellozia (Velloziaceae) Reveals Range Expansion in South American Mountaintops After Climatic Cooling Events and Increased Diversification Rates After the Occupation of Southern Espinhaço Province." *Botanical Journal of the Linnean Society* 207: 115–127. https://doi.org/10.1093/botlinnean/boae072.

Martini, A. M. Z., P. Fiaschi, A. M. Amorim, and J. L. Paixão. 2007. "A Hot-Point Within a Hot-Spot: A High Diversity Site in Brazil's Atlantic Forest." *Biodiversity and Conservation* 16: 3111–3128. https://doi.org/10.1007/s10531-007-9166-6.

Matzke, N. J. 2013. "Probabilistic Historical Biogeography: New Models for Founder-Event Speciation, Imperfect Detection, and Fossils Allow Improved Accuracy and Model-Testing." *Frontiers of Biogeography* 5, no. 4: 242–248. https://doi.org/10.21425/F5FBG19694.

Matzke, N. J. 2014. "Model Selection in Historical Biogeography Reveals That Founder-Event Speciation Is a Crucial Process in Island Clades." *Systematic Biology* 63: 951–970. https://doi.org/10.1093/sysbio/syu056.

McVay, J. D., and B. C. Carstens. 2013. "Testing Monophyly Without Well-Supported Gene Trees: Evidence from Multi-Locus Nuclear Data Conflicts with Existing Taxonomy in the Snake Tribe Thamnophiini." *Molecular Phylogenetics and Evolution* 68, no. 3: 425–431. https://doi.org/10.1016/j.ympev.2013.04.028.

Moro, M. F., V. O. Amorim, L. P. Queiroz, et al. 2024. "Biogeographical Districts of the Caatinga Dominion: A Proposal Based on Geomorphology and Endemism." *Botanical Review* 90, no. 4: 376–429. https://doi.org/10.1007/s12229-024-09304-5.

Moro, M. F., E. Nic Lughadha, F. S. Araújo, and F. R. Martins. 2016. "A Phytogeographical Metaanalysis of the Semiarid Caatinga Domain in Brazil." *Botanical Review* 82, no. 2: 91–148. https://doi.org/10.1007/s12229-016-9164-z.

Morrone, J. J., T. Escalante, G. Rodríguez-Tapia, A. Carmona, M. Arana, and J. D. Mercado-Gómez. 2022. "Biogeographic Regionalization of the Neotropical Region: New Map and Shapefile." *Anais da Academia Brasileira de Ciências* 94: e20211167.

Mota, G. S., G. R. Luz, N. M. Mota, et al. 2018. "Changes in Species Composition, Vegetation Structure, and Life Forms Along an Altitudinal Gradient of Rupestrian Grasslands in South-Eastern Brazil." *Flora* 238: 32–42. https://doi.org/10.1016/j.flora.2017.03.010.

Muellner-Riehl, A. N., J. Schnitzler, W. D. Kissling, et al. 2019. "Origins of Global Mountain Plant Biodiversity: Testing the Mountain-Geobiodiversity Hypothesis." *Journal of Biogeography* 46: 2826–2838. https://doi.org/10.1111/jbi.13715.

Müller, N. F., and R. R. Bouckaert. 2020. "Adaptive Metropolis-Coupled MCMC for BEAST 2." *PeerJ* 8: e9473. https://doi.org/10.7717/peerj.9473.

Myers, N., R. A. Mittermeier, C. G. Mittermeier, G. A. B. Fonseca, and J. Kent. 2000. "Biodiversity Hotspots for Conservation Priorities." *Nature* 403: 853–858. https://doi.org/10.1038/35002501.

Nery, E. K., M. K. Caddah, M. F. Santos, and A. Nogueira. 2023. "The Evolution of Ecological Specialization Underlies Plant Endemism in the Atlantic Forest." *Annals of Botany* 131: 921–940. https://doi.org/10.1093/aob/mcad029.

Neves, D. M., K. G. Dexter, R. T. Pennington, M. L. Bueno, P. L. S. Miranda, and A. T. Oliveira-Filho. 2018. "Lack of Floristic Identity in Campo



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Rupestre—A Hyperdiverse Mosaic of Rocky Montane Savannas in South America." *Flora* 238: 24–31. https://doi.org/10.1016/j.flora.2017.03.011.

Nguyen, L. T., H. A. Schmidt, A. von Haeseler, and B. Q. Minh. 2015. "IQ-TREE: A Fast and Effective Stochastic Algorithm for Estimating Maximum-Likelihood Phylogenies." *Molecular Biology and Evolution* 32: 268–274. https://doi.org/10.1093/molbev/msu300.

Nicholls, J. A., J. J. Ringelberg, K. G. Dexter, et al. 2025. "Continuous Colonization of the Atlantic Coastal Rain Forests of South America From Amazônia." *Proceedings of the Royal Society B: Biological Sciences* 292: 2039. https://doi.org/10.1098/rspb.2024.1559.

O'Brien, C. L., M. Huber, E. Thomas, et al. 2020. "The Enigma of Oligocene Climate and Global Surface Temperature Evolution." *Proceedings of the National Academy of Sciences* 117: 25302–25309. https://doi.org/10.1073/pnas.2003914117.

Oliveira, R. S., H. C. Galvão, M. C. R. Campos, C. B. Eller, S. J. Pearse, and H. Lambers. 2015. "Mineral Nutrition of Campo Rupestre Plant Species on Contrasting Nutrient-Impoverished Soil Types." *New Phytologist* 205: 1183–1194. https://doi.org/10.1111/nph.13175.

Pedreira, A. J., and B. de Waele. 2008. "Contemporaneous Evolution of the Palaeoproterozoic–Mesoproterozoic Sedimentary Basins of the São Francisco–Congo Craton." *Geological Society, London, Special Publications* 294: 33–48. https://doi.org/10.1144/SP294.3.

Pennington, R. T., J. E. Richardson, and M. Lavin. 2006. "Insights Into the Historical Construction of Species-Rich Biomes From Dated Plant Phylogenies, Neutral Ecological Theory and Phylogenetic Community Structure." *New Phytologist* 172: 605–616. https://doi.org/10.1111/j.1469-8137.2006.01902.x.

Pennington, T. R., D. E. Prado, and C. A. Pendry. 2000. "Neotropical Seasonally Dry Forests and Quaternary Vegetation Changes." *Journal of Biogeography* 27, no. 2: 261–273. https://doi.org/10.1046/j.1365-2699.2000.00397.x.

Pérez-Escobar, O. A., A. Zizka, M. A. Bermúdez, et al. 2022. "The Andes Through Time: Evolution and Distribution of Andean Floras." *Trends in Plant Science* 27: 364–378. https://doi.org/10.1016/j.tplants.2021.09.010.

Perrigo, A., C. Hoorn, and A. Antonelli. 2020. "Why Mountains Matter for Biodiversity." *Journal of Biogeography* 47: 315–325. https://doi.org/10.1111/jbi.13731.

Pires, R. C., F. B. Reis Junior, J. E. Zilli, et al. 2018. "Soil Characteristics Determine the Rhizobia in Association With Different Species of *Mimosa* in Central Brazil." *Plant and Soil* 423, no. 1–2: 411–428. https://doi.org/10.1007/s11104-017-3521-5.

Prance, G. T. 1982. "A Review of the Phytogeographic Evidences for Pleistocene Climate Changes in the Neotropics." *Annals of the Missouri Botanical Garden* 69: 594–624.

Prochazka, L. S., S. Alcantara, J. G. Rando, T. Vasconcelos, R. C. Pizzardo, and A. Nogueira. 2024. "Resource Availability and Disturbance Frequency Shape Evolution of Plant Life Forms in Neotropical Habitats." *New Phytologist* 242: 760–773. https://doi.org/10.1111/nph.19601.

Queiroz, L. P., and C. Snak. 2020. "Revisiting the Taxonomy of Dioclea and Related Genera (Leguminosae, Papilionoideae), with New Generic Circumscriptions." *PhytoKeys* 164: 67–114. https://doi.org/10.3897/phytokeys.164.55441.

Quesada, C. A., C. Paz, E. Oblitas Mendoza, O. L. Phillips, G. Saiz, and J. Lloyd. 2020. "Variations in Soil Chemical and Physical Properties Explain Basin-Wide Amazon Forest Soil Carbon Concentrations." *Soil* 6: 53–88. https://doi.org/10.5194/soil-6-53-2020.

Rahbek, C., M. K. Borregaard, R. K. Colwell, et al. 2019. "Humboldt's Enigma: What Causes Global Patterns of Mountain Biodiversity?" *Science* 365: 1108–1113. https://doi.org/10.1126/science.aax0149.

Rambaut, A., A. J. Drummond, D. Xie, G. Baele, and M. A. Suchard. 2018. "Posterior Summarization in Bayesian Phylogenetics Using Tracer 1.7." *Systematic Biology* 67: 901–904. https://doi.org/10.1093/sysbio/syy032.

Rando, J. G., M. M. T. Cota, A. S. Conceição, A. R. Barbosa, and T. L. A. Barros. 2020. "*Chamaecrista* in Flora do Brasil 2020." *Jardim Botânico Do Rio de Janeiro*. https://floradobrasil2020.jbrj.gov.br/FB22876.

Rando, J. G., M. M. T. Cota, A. Lima, R. L. C. Bortolozzi, B. Marazzi, and A. S. Conceição. 2024. "Tribe Cassieae." *PhytoKeys* 240: 83–102. https://doi.org/10.3897/phytokeys.240.101716.

Rando, J. G., A. R. Zuntini, A. S. Conceição, C. van den Berg, J. R. Pirani, and L. P. Queiroz. 2016. "Phylogeny of *Chamaecrista* Sect. *Chamaecrista* Ser. *Coricaeae* (Leguminosae) Unveils a Lineage Recently Diversified in Brazilian Campo Rupestre Vegetation." *International Journal of Plant Sciences* 177: 3–17. https://doi.org/10.1086/683846.

Rapini, A., C. Bitencourt, F. Luebert, and D. Cardoso. 2021. "An Escape-To-Radiate Model for Explaining the High Plant Diversity and Endemism in Campo Rupestre." *Biological Journal of the Linnean Society* 133: 481–498. https://doi.org/10.1093/biolinnean/blaa179.

Renner, S. 2004. "Plant Dispersal Across the Tropical Atlantic by Wind and Sea Currents." *International Journal of Plant Sciences* 165: S23–S33. https://doi.org/10.1086/383334.

Robinson, D. F., and L. R. Foulds. 1981. "Comparison of Phylogenetic Trees." *Mathematical Biosciences* 53: 131–147. https://doi.org/10.1016/0025-5564(81)90043-2.

Rouws, L., A. Barauna, C. Beukes, et al. 2024. "Soil Characteristics Drive Contrasting Patterns of Association Between Symbiotic Rhizobia of Endemic and Widespread *Mimosa* Species in Brazil." *Applied Soil Ecology* 204: 105741. https://doi.org/10.1016/j.apsoil.2024.105741.

Rutter, M. T., and M. D. Rausher. 2004. "Natural Selection on Extrafloral Nectar Production in *Chamaecrista fasciculata*: The Costs and Benefits of a Mutualism Trait." *Evolution* 58: 2657–2668. https://doi.org/10.1111/j.0014-3820.2004.tb01619.x.

Santos, M. F., E. Lucas, P. T. Sano, S. Buerki, V. G. Staggemeier, and F. Forest. 2017. "Biogeographical Patterns of *Myrcia* s.l. (Myrtaceae) and Their Correlation With Geological and Climatic History in the Neotropics." *Molecular Phylogenetics and Evolution* 108: 34–48. https://doi.org/10.1016/j.ympev.2017.01.012.

Santos, R. M., A. T. Oliveira-Filho, P. V. Eisenlohr, L. P. Queiroz, D. B. O. S. Cardoso, and M. J. N. Rodal. 2012. "Identity and Relationships of the Arboreal Caatinga Among Other Floristic Units of Seasonally Dry Tropical Forests (SDTFs) of North-Eastern and Central Brazil." *Ecology and Evolution* 2, no. 2: 409–428. https://doi.org/10.1002/ece3.91.

Saura, S., Ö. Bodin, and M. Fortin. 2014. "Stepping Stones Are Crucial for Species' Long-Distance Dispersal and Range Expansion Through Habitat Networks." *Journal of Applied Ecology* 51, no. 1: 171–182. https://doi.org/10.1111/1365-2664.12179.

Schaefer, C. E., G. R. Corrêa, H. G. Candido, et al. 2016. "The Physical Environment of Rupestrian Grasslands (Campos Rupestres) in Brazil: Geological, Geomorphological and Pedological Characteristics, and Interplays." In *Ecology and Conservation of Mountaintop Grasslands in Brazil*, edited by G. Fernandes. Springer. https://doi.org/10.1007/978-3-319-29808-5_2.

Schliep, K., A. J. Potts, D. A. Morrison, and G. W. Grimm. 2017. "Intertwining Phylogenetic Trees and Networks." *Methods in Ecology and Evolution* 8: 1212–1220. https://doi.org/10.1111/2041-210X.12760.

Silberbauer-Gottsberger, G. G. 2006. "Life in the Cerrado: A South American Tropical Seasonal Vegetation." In *Origin, Structure, Dynamics and Plant Use*, vol. I. Reta Verlag.



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Silva, A. C., and A. F. Souza. 2018. "Aridity Drives Plant Biogeographical Sub Regions in the Caatinga, the Largest Tropical Dry Forest and Woodland Block in South America." *PLoS One* 13, no. 4: e0196130. https://doi.org/10.1371/journal.pone.0196130.

Silva, M. C., L. Rowland, R. S. Oliveira, R. T. Pennington, and P. Moonlight. 2024. "Elevation Modulates the Impacts of Climate Change on the Brazilian Cerrado Flora." *Diversity and Distributions* 30, no. 5: e13832. https://doi.org/10.1111/ddi.13832.

Silveira, F. A. O., D. Negreiros, N. P. U. Barbosa, et al. 2016. "Ecology and Evolution of Plant Diversity in the Endangered *Campo Rupestre*: A Neglected Conservation Priority." *Plant and Soil* 403: 129–152. https://doi.org/10.1007/s11104-015-2637-8.

Simon, M. F., R. Grether, L. P. Queiroz, C. Skema, R. T. Pennington, and C. E. Hughes. 2009. "Recent Assembly of the Cerrado, a Neotropical Plant Diversity Hotspot, by In Situ Evolution of Adaptations to Fire." *Proceedings of the National Academy of Sciences* 106: 20359–20364. https://doi.org/10.1073/pnas.0903410106.

Simon, M. F., and T. Pennington. 2012. "Evidence for Adaptation to Fire Regimes in the Tropical Savannas of the Brazilian Cerrado." *International Journal of Plant Sciences* 173: 711–723. https://doi.org/10.1086/665973.

Souza, A. O., G. P. Lewis, and M. J. Silva. 2021. "A New Infrageneric Classification of the Pantropical Genus *Chamaecrista* (Fabaceae: Caesalpinioideae) Based on a Comprehensive Molecular Phylogenetic Analysis and Morphology." *Botanical Journal of the Linnean Society* 197: 350–395. https://doi.org/10.1093/botlinnean/boab029.

Souza, C., L. C. Leal, V. Lima, et al. 2025. "High Dependence on Nitrogen-Fixing Bacteria in a Neotropical Legume Species." *Symbiosis* 95, no. 2: 215–224. https://doi.org/10.1007/s13199-025-01042-5.

Souza, C., L. B. Valadão-Mendes, I. Schulze-Albuquerque, P. J. Bergamo, D. D. Souza, and A. Nogueira. 2024. "Nitrogen-Fixing Bacteria Boost Floral Attractiveness in a Tropical Legume Species During Nutrient Limitation." *American Journal of Botany* 111, no. 12: e16363. https://doi.org/10.1002/ajb2.16363.

Steinthorsdottir, M., H. K. Coxall, A. M. de Boer, et al. 2021. "The Miocene: The Future of the Past." *Paleoceanography and Paleoclimatology* 36: e2020PA004037. https://doi.org/10.1029/2020PA004037.

Strömberg, C. A. E. 2011. "Evolution of Grasses and Grassland Ecosystems." *Annual Review of Earth and Planetary Sciences* 39: 517–544. https://doi.org/10.1146/annurev-earth-040809-152402.

Tuomisto, H., G. M. Moulatlet, H. Balslev, et al. 2016. "A Compositional Turnover Zone of Biogeographical Magnitude Within Lowland Amazonia." *Journal of Biogeography* 43: 2400–2411. https://doi.org/10.1111/jbi.12864.

Tuomisto, H., K. Ruokolainen, R. Kalliola, A. Linna, W. Danjoy, and Z. Rodriguez. 1995. "Dissecting Amazonian Biodiversity." *Science* 269: 63–66. https://doi.org/10.1126/science.269.5220.63.

Vasconcelos, T. N. C., S. Alcantara, C. O. Andrino, M. Simon, and J. R. Pirani. 2020. "Fast Diversification Through a Mosaic of Evolutionary Histories Characterizes the Endemic Flora of Ancient Neotropical Mountains." *Proceedings of the Royal Society B: Biological Sciences* 287: 20192933. https://doi.org/10.1098/rspb.2019.2933.

Vidal, J. D., A. P. Souza, and I. Koch. 2019. "Impacts of Landscape Composition, Marginality of Distribution, Soil Fertility and Climatic Stability on the Patterns of Woody Plant Endemism in the Cerrado." *Global Ecology and Biogeography* 28: 904–916. https://doi.org/10.1111/geb.12901.

Vilhena, D., and A. Antonelli. 2015. "A Network Approach for Identifying and Delimiting Biogeographical Regions." *Nature Communications* 6: 6848. https://doi.org/10.1038/ncomms7848.

Vivo, M. d. 1997. "Mammalian Evidence of Historical Ecological Change in the Caatinga Semiarid Vegetation of Northeastern Brazil." *Journal of Comparative Biology* 2: 65–73.

Werneck, F. P. 2011. "The Diversification of Eastern South American Open Vegetation Biomes: Historical Biogeography and Perspectives." *Quaternary Science Reviews* 30: 1630–1648. https://doi.org/10.1016/j.quascirev.2011.03.00.

Zizka, A., D. Silvestro, T. Andermann, et al., 2019. "CoordinateCleaner: Standardized Cleaning of Occurrence Records from Biological Collection Databases." *Methods in Ecology and Evolution* 10: 744–751. https://doi.org/10.1111/2041-210X.13152.

## Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Material and Methods. **Data S2:** jbi70137-sup-0002-Supporting_Information S2.tiff. **Data S3:** Bioregions for *Chamaecrista* inferred using a network-based approach implemented in the Infomap Bioregions interactive web application. **Data S4:** Distribution of the species, coded according to the biogeographic areas adopted in the study. **Data S5:** Matrix of dispersal rates across time frames. **Data S6:** Estimated dispersal rates from biogeographic reconstruction.



13652699, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jbi.70137 by Capes, Wiley Online Library on [14/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License